Introduction to Photogrammetry

Lecture 16

Stereo Geometry

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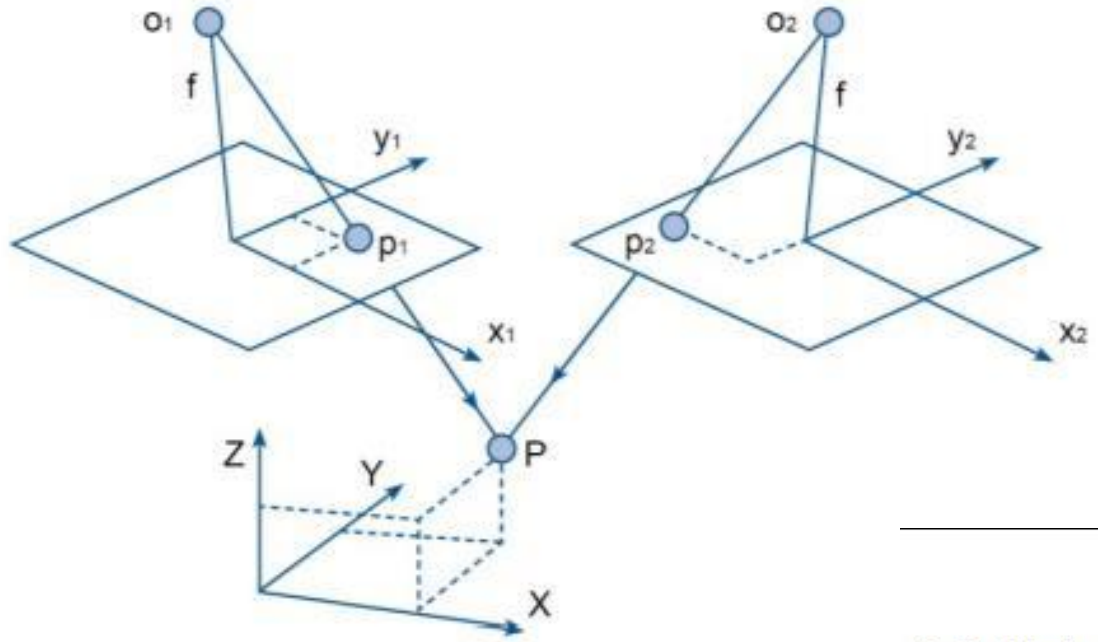
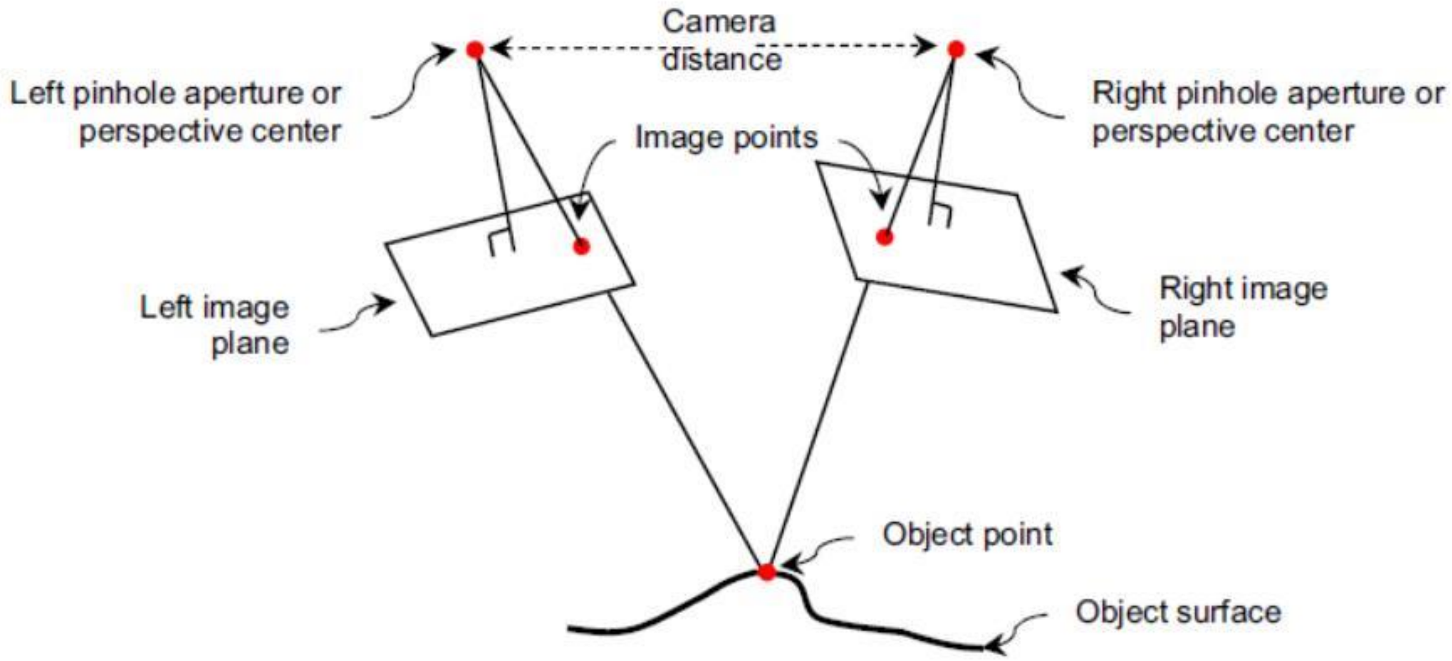
In Today's Class

1. Introduction
2. Parallax in Photogrammetry

1. Introduction

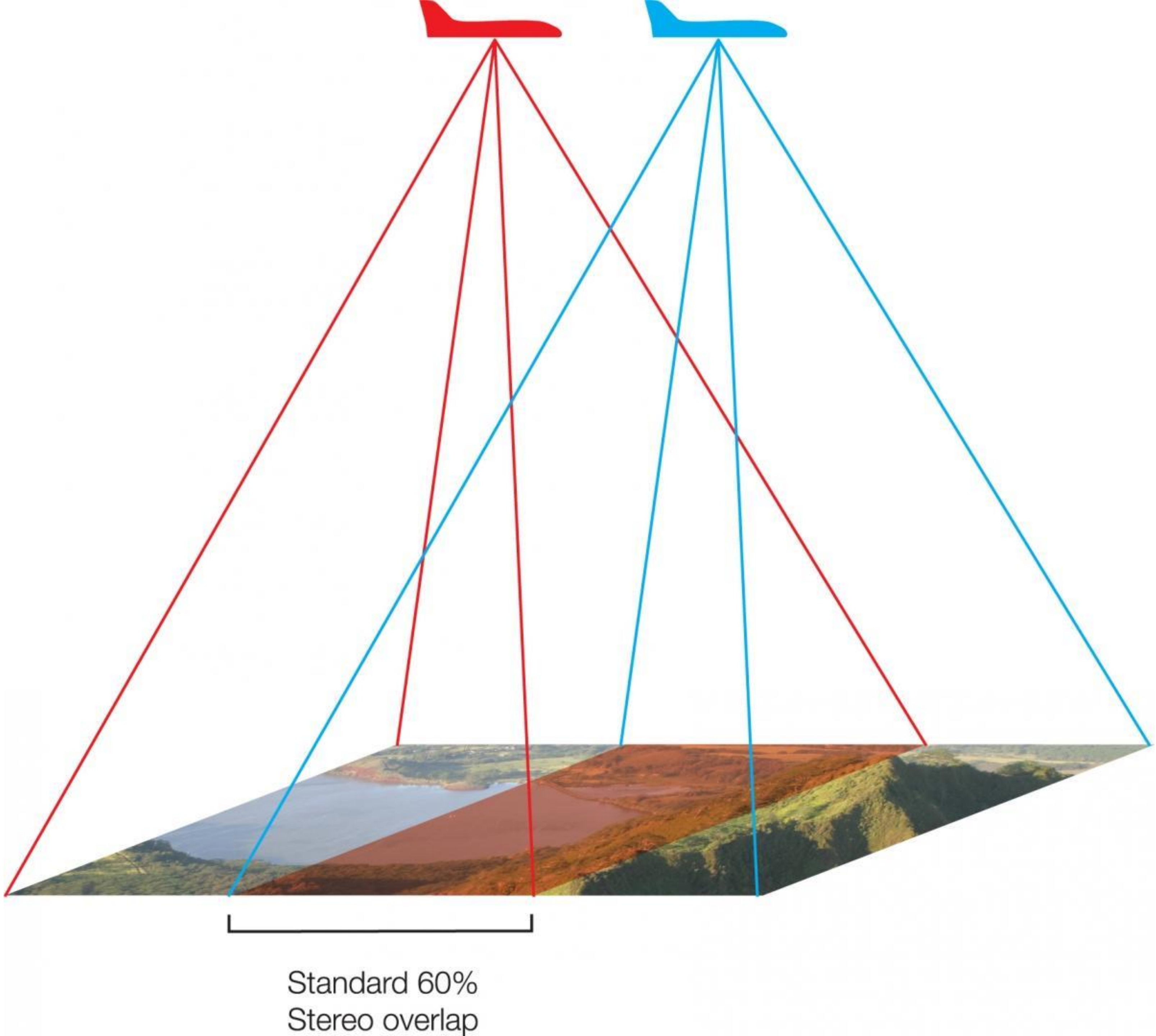
- Human vision perceives depth because each eye observes objects from slightly different positions.
- Photogrammetry uses overlapping photographs to create a similar stereoscopic effect.
- Stereo vision allows terrain elevation and shape to be interpreted visually.
- The perception of depth is essential for three-dimensional mapping.


o1
o2
f
f
y1
y2
p1
p2
x1
x2
Z
Y
P
X


Camera distance
Left pinhole aperture or perspective center
Right pinhole aperture or perspective center
Image points
Left image plane
Right image plane
Object point
Object surface

Stereo Pair in Photogrammetry

- A stereo pair consists of two overlapping aerial photographs of the same area.
- The overlap between photographs is usually around sixty percent.
- Common points appearing in both photographs are used for stereo reconstruction.
- Stereo pairs are essential for creating three-dimensional terrain models.

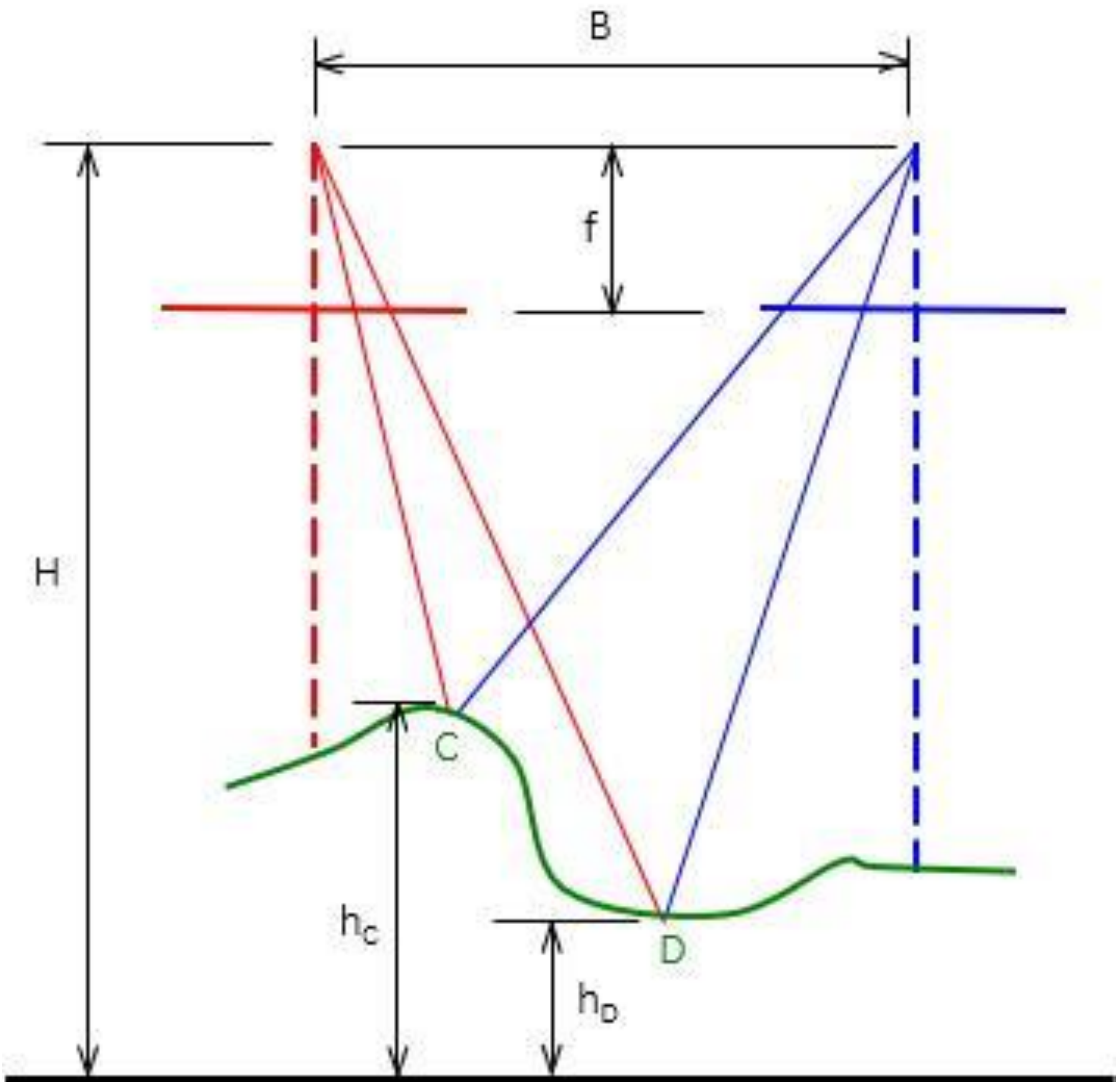

Standard 60%
Stereo overlap

Stereoscopic Model

- A stereoscopic model is formed when two oriented photographs are viewed together.
- The stereo model creates the illusion of three-dimensional terrain.
- Heights and terrain features can be measured from the stereo model.
- Accurate orientation is required for reliable stereo reconstruction.

2. Parallax in Photogrammetry

- Parallax is the apparent displacement of an object between overlapping photographs.
- Objects at different elevations show different amounts of parallax.
- Parallax is the fundamental basis for height determination in photogrammetry.
- Greater elevation differences produce larger parallax differences.


B
f
H
C
D
h_C
h_D

X-Parallax and Y-Parallax

- X-parallax occurs along the flight direction of aerial photographs.
- Y-parallax occurs perpendicular to the flight direction.
- Relative orientation aims to eliminate y-parallax from stereo images.
- Residual y-parallax indicates orientation errors.

Importance of Removing Y-Parallax

- Y-parallax prevents proper stereoscopic viewing of photographs.
- Elimination of y-parallax improves three-dimensional image fusion.
- Accurate relative orientation depends on minimizing y-parallax.
- Proper orientation produces a stable stereo model.

Thank you!

Any Questions